DIGITAL TWIN–EINSTEIN TELESCOPE (DT–ET) XR RESEARCH PROJECTS

Project 08: Surfing a Gravitational Wave

Travel with a ripple in spacetime and investigate what changes when your perspective changes.

Year 2 – Unity + XR – Team of 5 students

Research line: Digital Twin–Einstein Telescope, Data Intelligence Research Centre, Zuyd University of Applied Sciences

Your mission. Build an interactive XR Digital Twin research prototype that lets a user observe a simplified gravitational wave from different spatial perspectives – including a deliberately slowed "ride-with-the-wave" experience. Users place test objects, pause and replay the wave, inspect stretching/squeezing, and investigate how immersive perspective can support spatial understanding of gravitational-wave propagation.

You are not expected to be a physicist

Part of the challenge is learning enough physics to represent a gravitational wave responsibly. The Starter Pack introduces propagation, strain, transverse stretching/squeezing and test-object visualizations. You do **not** need general-relativity derivations. Your prototype should be scientifically meaningful, explicitly simplified and designed for investigation.

1. DT–ET Research Context

NASA describes gravitational waves as invisible ripples in spacetime that propagate at the speed of light and stretch and squeeze distances as they pass [1]. LIGO's educational visualization deliberately exaggerates this minute deformation so it can be seen, while its educator guide warns that visual analogies must not be interpreted too literally [2, 3]. The Einstein Telescope will detect gravitational waves through extremely small relative length changes in long laser-interferometer arms [4].

This project therefore uses XR as an **impossible laboratory**: the real effect is far too small and fast for a human to see directly, but a transparent Digital Twin can slow time, exaggerate deformation and let the user change viewpoint. Every exaggeration must be labelled as a visualization choice rather than a claim about physical scale.

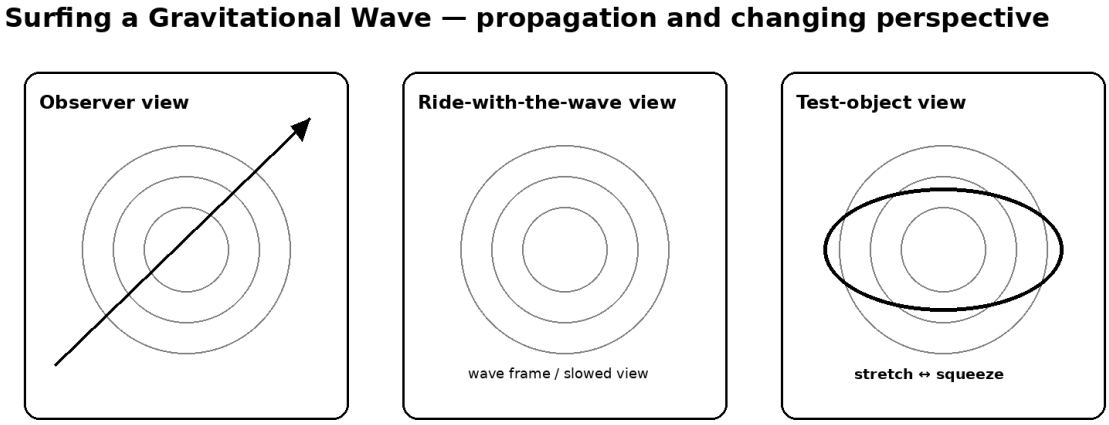


Figure 1: Original course illustration: the same simplified phenomenon can be investigated from observer, ride-with-the-wave and test-object perspectives.

2. The Challenge

Project 01 asks primarily: *what happens when a gravitational wave reaches an observer/detector?* Project 08 asks a different question: *how can we understand propagation by moving our viewpoint with respect to the wave?* The user should encounter a wavefront, slow or freeze the simulation, travel alongside a deliberately slowed representation, place test objects at different positions, and compare how distances change as the wave passes.

Core experience: Observe → slow/pause → change viewpoint → place test objects → replay → compare spatial effects → explain what the visualization means physically.

3. Physics Starter Pack

3.1 Propagation and strain

Gravitational waves propagate through spacetime at the speed of light [1]. A useful quantity is dimensionless strain,

$$h = \frac{\Delta L}{L}, \tag{1}$$

which expresses a relative change in separation. For this Year-2 prototype, students may use a pedagogical sinusoidal state such as

$$h(t) = h_0 \sin(2\pi f t + \phi), \tag{2}$$

then map that state to an exaggerated deformation of a ring or grid of freely positioned test objects. This is a teaching model, not a full solution of Einstein's field equations.

3.2 Stretch and squeeze

A passing gravitational wave can produce alternating stretching in one transverse direction and squeezing in the perpendicular direction; LIGO and NASA use this visual language in public and educational explanations [2, 1]. A ring of test particles is therefore a useful conceptual visualization.

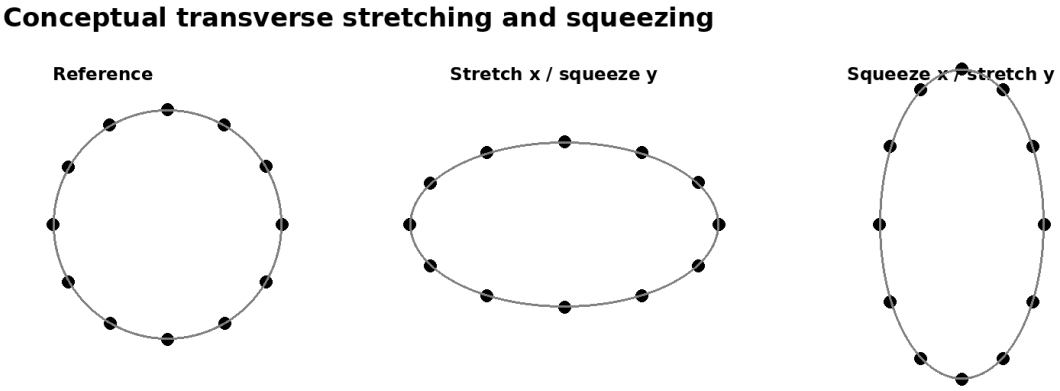


Figure 2: Original conceptual illustration of alternating transverse deformation. The deformation shown is enormously exaggerated for visibility, following the pedagogical principle used in LIGO's public visualization [2].

3.3 A crucial XR simplification

In reality, a user cannot literally "surf" next to a gravitational wave as portrayed in the experience: the wave propagates at light speed. The XR system may intentionally reduce the displayed propagation speed or use a moving reference frame so the process can be inspected. The interface must explicitly say **simulation speed / visualization speed**, not imply that the real wave moves slowly.

Scope control

You do not need to model: Einstein's field equations, tensor calculus, numerical relativity, realistic detector sensitivity, astrophysical waveform generation, curved-spacetime ray tracing, or relativistic observer transformations. Do not claim the XR "ride" is a literal physical journey.

4. From Physics to Digital Twin

The prototype must preserve a visible chain from physical idea to model state. Parameters such as h_0, f, ϕ, propagation direction, simulation time, visualization exaggeration and viewpoint should update the Digital Twin state and then the XR scene.

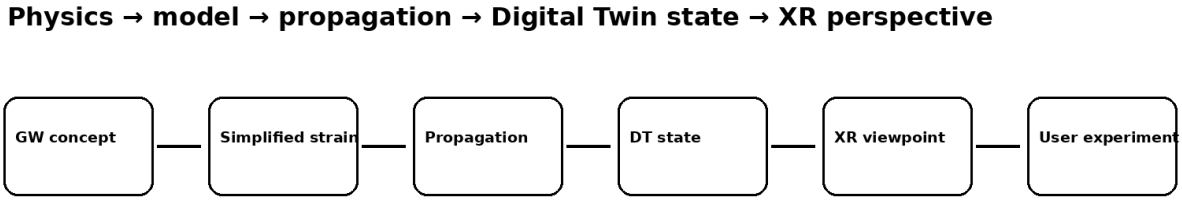


Figure 3: Required model-driven loop for the research prototype.

5. XR Experience

A strong experience begins in an observer frame with a reference grid or field of test objects. A wave approaches and the user sees an intentionally exaggerated change in relative separations. The user

then pauses the wave, walks around the field, changes amplitude/frequency within the teaching model and replays the event.

The distinctive feature is a **perspective mode**. The user can switch to a slowed propagation view or a camera/reference frame that travels with the visualization. Markers placed ahead of and behind the current wavefront help the user reason about where the disturbance has reached. A comparison mode should return to the stationary observer and ask the user to explain what was invariant and what changed only because of visualization or viewpoint.

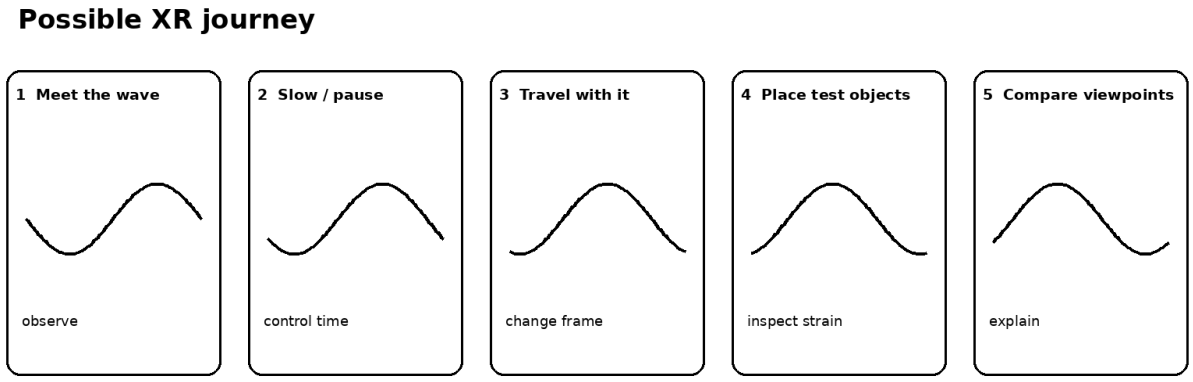


Figure 4: Possible XR journey. The ride-with-the-wave stage is explicitly a pedagogical visualization, not a literal relativistic observer.

6. Research Component

The prototype is a research artifact, not the research conclusion.

> **Suggested main research question:** How can immersive XR visualization support spatial understanding of gravitational-wave propagation and its effects on physical distances?

Possible focused comparisons include stationary observer versus perspective-changing experience; animated grid versus test-particle ring; normal playback versus pause/slow/replay; or a single viewpoint versus multiple viewpoints. One well-defined comparison is enough.

7. Minimum Viable Prototype (MVP)

A successful MVP contains:

- a functional Unity XR scene with a propagating gravitational-wave representation;
- a transparent simplified strain model;
- a grid, ring or field of test objects that visibly responds;
- play, pause, slow/replay and reset;
- at least two meaningful parameters such as amplitude and frequency;
- at least two spatial viewpoints, including a stationary observer and a propagation/perspective mode;
- explicit labels for exaggerated deformation and altered visualization speed;
- a focused research question and small evaluation.

Optional extensions include two polarization visualizations, multiple test-object geometries, measurement tools, guided prediction tasks, comparison with an interferometer-arm representation, or a bridge to an astrophysical source. Extensions come only after the MVP is stable.

8. Research and Evaluation

Evaluation should test spatial understanding rather than entertainment. Example tasks: predict which test objects will be affected next; identify stretch versus squeeze; explain why the displayed deformation is exaggerated; distinguish physical parameters from visualization controls; or reconstruct the propagation direction from observations at several positions. Evidence can include task accuracy, short pre/post concept questions, completion time, confidence, usability and brief interviews.

9. Expected Deliverables

Deliver a working Unity/XR prototype; concise physics/model documentation; a clear table of physical assumptions versus visualization choices; research question and evaluation method; evidence and analysis; conclusions and limitations; and a final demonstration that makes the chain **Gravitational-wave concept** → **simplified model** → **propagation** → **Digital Twin state** → **XR perspective** → **research** explicit.

10. Milestones

Period	Milestone	Expected progress
Week 2	Project Kick-off (planned for Friday)	DT–ET context, Starter Pack, first literature search, choose test-object representation and scope.
Weeks 3–5	Research & prototype development	Understand propagation/strain, define research question, implement simplified model, first Unity/XR observer view and perspective mode.
Week 6	Concept Pitch (planned for Friday)	Show early prototype; explain "The Physics Behind Our Digital Twin" in about two minutes; identify every exaggeration/simplification and present evaluation plan.
Weeks 7–9	Development, experiment & analysis	Complete interactions and comparison modes, conduct evaluation, analyse evidence and improve prototype where justified.
Week 10	Final Demo (planned for Friday)	Demonstrate the XR Digital Twin and present physics, model, research method, findings, limitations and conclusions.

11. Authoritative Starting Websites

- **NASA Space Place – What Is a Gravitational Wave?** https://spaceplace.nasa.gov/gravitational-waves/en/ – accessible explanation of propagation, stretching/squeezing and detection [1].
- **LIGO/Caltech – Exaggerated Effects of Gravitational Waves on Earth.** https://www.ligo.caltech.edu/MIT/video/ligo20160211v5 – especially useful for discussing why deformation must be exaggerated visually [2].
- **LIGO/Caltech – Educator's Guide.** https://www.ligo.caltech.edu/system/media_files/binaries/303/original/ligo-educators-guide.pdf – accessible physics plus important cautions about spacetime visualizations [3].

- **Einstein Telescope EMR – Optics and optical metrology.** https://www.einsteintelescope-emr.eu/en/optics-and-optical-metrology/ – connection between gravitational-wave strain and the ET measurement concept [4].
- **NASA Hubble – Gravity Waves.** https://science.nasa.gov/mission/hubble/science/science-behind-the-discoveries/hubble-gravity-waves/ – additional scientific overview [5].

12. Scientific References

- B. P. Abbott et al. (LIGO Scientific Collaboration and Virgo Collaboration), "Observation of Gravitational Waves from a Binary Black Hole Merger," *Physical Review Letters* 116, 061102 (2016). DOI: 10.1103/PhysRevLett.116.061102.
- M. Punturo et al., "The Einstein Telescope: a third-generation gravitational wave observatory," *Classical and Quantum Gravity* 27, 194002 (2010). DOI: 10.1088/0264-9381/27/19/194002.
- M. Maggiore et al., "Science Case for the Einstein Telescope," *JCAP* 2020(03), 050 (2020). DOI: 10.1088/1475-7516/2020/03/050.

References

References

[1] NASA Space Place, "What Is a Gravitational Wave?" https://spaceplace.nasa.gov/gravitational-waves/en/.

[2] LIGO Laboratory, Caltech/MIT, "Exaggerated Effects of Gravitational Waves on Earth," 2016. https://www.ligo.caltech.edu/MIT/video/ligo20160211v5.

[3] LIGO Scientific Collaboration and Virgo Collaboration, *Direct Observation of Gravitational Waves: Educator's Guide*, LIGO-P1600015-v6 (2016). https://www.ligo.caltech.edu/system/media_files/binaries/303/original/ligo-educators-guide.pdf.

[4] Einstein Telescope EMR, "Optics and optical metrology." https://www.einsteintelescope-emr.eu/en/optics-and-optical-metrology/.

[5] NASA Science, "Hubble Gravity Waves." https://science.nasa.gov/mission/hubble/science/science-behind-the-discoveries/hubble-gravity-waves/.

[6] B. P. Abbott et al., *Phys. Rev. Lett.* 116, 061102 (2016). https://doi.org/10.1103/PhysRevLett.116.061102.

[7] M. Punturo et al., *Class. Quantum Grav.* 27, 194002 (2010). https://doi.org/10.1088/0264-9381/27/19/194002.

[8] M. Maggiore et al., *JCAP* 2020(03), 050 (2020). https://doi.org/10.1088/1475-7516/2020/03/050.